Sweetspots

Full Cost Driver & Unit Economics Analysis

v2 — Updated with Routes API optimization & Discovery-First model

1. Cost Drivers: Every External API Call

A. Google Maps Platform

API	Price/call	Where used	When triggered
Places Text Search	$0.032	unified-search, generate-trip	Every vibe search (1-2 pages)
Places Autocomplete	$0.00283	place-autocomplete	Destination search keystrokes
Place Details	$0.025	enrich-places	User taps a place card
Place Photos	$0.007	place-photo	Every photo load (cached after first)
Geocoding	$0.005	unified-search	Every search with city name
~~Routes API~~	~~$0.01/ route~~	~~unified-search~~	**REMOVED — replaced with Haversine**
Routes API	$0.01/ route	compute-routes	Trip distance connectors only
Street View Static	$0.007	street-view	Place detail page

B. Lovable AI Gateway

Model	Est. cost/call	Where used	When triggered
gemini-2.5-flash-lite	$0.001–0.003	unified-search	Prompt translation + tag gen
gemini-2.5-flash	$0.003–0.01	unified-search, enrich	AI scoring, enrichment
gemini-3-flash-preview	$0.005–0.015	generate-trip, import	Trips, imports, swaps

C. Infrastructure (Lovable Cloud)

Resource	Cost driver	Notes
Database (Postgres)	Storage + compute	Grows with places + cache
Edge Functions	Invocations	Each call = invocation
Storage (photos)	GB stored + bandwidth	Photos cached permanently

2. Cost Per User Action

✔ **Routes API removed from search — 71% cost reduction per search**

Vibe Search (core action) — OPTIMIZED

Step	API calls	Est. cost
Prompt translation (AI)	1× gemini-flash-lite	$0.002
Google Places Text Search	2× searchText	$0.064
Google Geocoding	1× geocode	$0.005
AI Relevance Scoring	1× gemini-flash	$0.006
AI Filter Tags	1× gemini-flash-lite	$0.002
~~Google Routes (old)~~	~~20× routes~~	**$0.20 → $0.00**
Haversine (local calc)	0 API calls	$0.00
TOTAL (uncached)		**$0.08 (was $0.28)**

Place Detail View

Step	API calls	Est. cost
Photo load (cache miss)	1–3× place-photo	$0.007–$0.021
Enrichment (if needed)	1× Details + 2× AI	$0.04
Street View	1× metadata + 1× image	$0.007
Similar Vibes	1× gemini-flash	$0.006
TOTAL (first view)		**$0.07**

Generate a Trip

Step	API calls	Est. cost
Parse trip prompt	1× gemini-flash	$0.008
Generate itinerary	1× gemini-flash	$0.015
Enrich activities	~9× searchText	$0.288
TOTAL		**$0.31**

Other Actions

Action	Cost	Notes

Swap Trip Activity	$0.008	1× AI call
Import from Link	$0.04	AI + 1× searchText
Character Match	$0.008	1× AI call
Board Suggestions	$0.09 (was $0.29)	AI + unified-search
Destination Autocomplete	$0.014/session	~5 keystrokes

3. Monthly Cost: Two Models Compared

Model A: Template-First (Current)

Users browse pre-made trip templates on the homepage.

Action	Freq/mo	Cost/action	Avg cost	Monthly
Vibe searches	15	$0.08	50% cached	$0.60
Place detail views	30	$0.07	70% cached	$0.63
Trip generation	2	$0.31	—	$0.62
Link imports	5	$0.04	—	$0.20
Activity swaps	3	$0.008	—	$0.024
Destination search	4	$0.014	—	$0.056
Board suggestions	1	$0.09	—	$0.09
Character match	1	$0.008	—	$0.008
TOTAL				**$2.23/user/mo**

✔ **Was $3.93 before Routes optimization — 43% reduction**

Model B: Discovery-First

Users search → browse results → view details → save → build trips.

Action	Freq/mo	Cost/action	Avg cost	Monthly
Vibe searches	35	$0.08	40% cached	$1.68
Place detail views	60	$0.07	60% cached	$1.68
Trip generation	2	$0.31	—	$0.62
Link imports	3	$0.04	—	$0.12
Activity swaps	4	$0.008	—	$0.032
Destination search	6	$0.014	—	$0.084
Board suggestions	2	$0.09	—	$0.18
Character match	1	$0.008	—	$0.008
TOTAL				**$4.40/user/mo**

⚠ Discovery-first costs ~2× template-first, but Routes optimization keeps it viable.

Side-by-Side Summary

Metric	Template-First	Discovery-First

Variable cost/user/month	$2.23	$4.40
Break-even subscription	~$2.50/mo	~$5/mo
Recommended Pro price	$4.99/mo	$7.99/mo
Gross margin at Pro	55%	45%
Biggest cost driver	Place details (28%)	Vibe searches (38%)

4. Business Model Recommendations

Option A: Freemium Subscription (Recommended)

Template-First Pricing

Tier	Price	Included	COGS	Margin
Free	$0	3 searches/day, 1 trip/mo	$0.30	—
Pro	$4.99/mo	Unlimited searches, 5 trips/mo	$2.23	55%
Pro+	$9.99/mo	Collaborative trips, priority AI	$3.50	65%

Discovery-First Pricing

Tier	Price	Included	COGS	Margin
Free	$0	5 searches/day, 1 trip/mo	$0.70	—
Pro	$7.99/mo	Unlimited searches, 5 trips/mo	$4.40	45%
Pro+	$14.99/mo	Collaborative trips, priority AI	$6.50	57%

Option B: Credit/Token System

Action	Credits	Your cost	Notes
1 vibe search	3	$0.08	Was 5 credits pre-optimization
Generate trip	10	$0.31	Most expensive action
Import link	1	$0.04	Good margin
Place detail view	1	$0.07	Break-even
Board suggestions	3	$0.09	Similar to search

Option C: Hybrid

Free base app with limited searches/trips + Pro subscription ($5.99–$7.99/mo) + extra trip credits at $0.50/trip beyond Pro limits.

5. Optimization Tracker

#	Optimization	Status	Impact
1	Replace Routes API with Haversine	☐ DONE	Saved $0.20/search (71%)
2	Photo caching	☐ Live	Cache hit rate >80%
3	Search result caching (7-day TTL)	☐ Live	Repeat searches = $0
4	Geocode caching (30-day TTL)	☐ Live	Minimal cost anyway
5	Reduce Places pages to 1	☐ TODO	Save $0.032/search
6	Lazy enrichment on tap only	☐ TODO	Save $0.04/place not viewed
7	Use flash-lite everywhere	☐ TODO	60% AI cost savings
8	Smaller photo thumbnails	☐ TODO	50% photo cost savings

6. Key Metrics to Track

Metric	Why it matters
Searches/user/day	Directly drives Google API costs
Cache hit rate (search)	Higher = lower marginal cost
Photo cache hit rate	Lower storage bandwidth + Google costs
Trips generated/user/month	Highest single-action cost
Places enriched (cumulative)	Enrichment amortizes over time
API cost/user/month (blended)	Your north star unit economics metric

TL;DR

✔ Routes API removed from search — biggest optimization done

Template-first cost: ~$2.23/user/month (was $3.93 → 43% savings)

Discovery-first cost: ~$4.40/user/month (viable with optimization)

Recommended Pricing

Template-first → $4.99/mo Pro (55% margin)

Discovery-first → $7.99/mo Pro (45% margin)

Next biggest wins: reduce Places pages, lazy enrichment, smaller thumbnails.